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From the Combinatonal model to Constructible Sheaves

Last time The six-functor formalism on locally compact Haus. spaces, convolution products

Today Constructing the comparison functor

$$\Psi_\sigma : (\mathrm{Fun}(\Theta(\sigma)^{op}, Sp), \otimes_{Day}) \longrightarrow (Sh(M_{\mathbb{R}}), *).$$

Recall If $M \in CMon(LCH)$ with addition $a: M \times M \to M$ and zero $0: * \to M$, then $Sh(M)$ has a presentably symmetric monoidal structure $*$ called the **convolution product** with

$$F * G \simeq a_!(pr_1^*(F) \otimes pr_2^*(G))$$

and unit the skyscraper sheaf $0_!(\mathbb{S}) = 0_*(\mathbb{S})$. Moreover, the dualizing sheaf $\omega_M \in Sh(M)$ is naturally a commutative algebra for convolution.

The convolution product of a real vector space

Notation If X is a topological space, $j: U \hookrightarrow X$ is an open subspace, and $F \in Sh(U)$, we often write $F \in Sh(X)$ for $j_!(F)$. In particular, we'll write $\mathbb{1}_U \in Sh(X)$ for $j_!(\mathbb{1}_U)$.
Similarly, if $i: Z \hookrightarrow X$ is closed and $F \in Sh(Z)$, we write $F \in Sh(X)$ for $i_*(F)$

Prop Let V be a finite dimensional real vector space and $X, Y \subset V$ polyhedral open subsets. Then there is a natural equivalence

$$\mathbb{1}_X * \mathbb{1}_Y \simeq \mathbb{1}_{X+Y}[-\dim(V)].$$

Here, $X+Y$ denotes the Minkowski sum

> The proof needs the following general result.

Prop (recognition criterion for locally constant sheaves) Let X be a locally weakly contractible topological space. Then a hypersheaf $L \in Sh^{hyp}(X)$ is locally constant if and only if for every inclusion of opens $U \hookrightarrow V$ that is a weak homotopy equivalence, the restriction map

$$L(V) \longrightarrow L(U)$$

is an equivalence.
If X is also weakly contractible, then every locally constant hypersheaf is constant.

Proof of convolution formula
Let $a: V \times V \longrightarrow V$ denote addition, and note that $X+Y$ is the image of

$$X \times Y \hookrightarrow V \times V \xrightarrow{a} V.$$

Let $j: X \hookrightarrow V$ and $j': Y \hookrightarrow V$ be the inclusions. By definition

$$\mathbb{1}_X * \mathbb{1}_Y = a_!\left(pr_1^*(j_!(\mathbb{1}_X)) \otimes pr_2^*(j'_!(\mathbb{1}_Y))\right)$$

By open basechange, we see that

$$\begin{aligned} pr_1^*(j_!(\mathbb{1}_X)) &\simeq (j \times id)_!(pr_1^*(\mathbb{1}_X)) \\ &\simeq (j \times id)_!(\mathbb{1}_{X \times V}) \end{aligned}$$

Similarly,

$$pr_2^*(j'_!(\mathbb{1}_Y)) \simeq (id \times j')_!(\mathbb{1}_{V \times Y})$$

We deduce that

$$\begin{aligned} pr_1^*(j_!(\mathbb{1}_X)) \otimes pr_2^*(j'_!(\mathbb{1}_Y)) &\simeq (j \times j')_!(\mathbb{1}_{X \times Y}) \\ &\simeq \mathbb{1}_{X \times Y} \end{aligned}$$

Hence

$$\mathbb{1}_X * \mathbb{1}_Y \simeq (a|_{X \times Y})_!(\mathbb{1}_{X \times Y}).$$

It is not hard to show that $a|_{X\times Y} : X\times Y \longrightarrow X+Y$ is a smooth V-bundle. By Poincaré duality, the $!$-pushforward along addition is locally constant. Since $X+Y$ is contractible, by the prev prop it is constant. Thus the claim reduces to the fact that for any $Z\in LCH$, for the projection $p: Z\times V \longrightarrow Z$ we have

$$p_!(\mathbb{1}) \simeq \mathbb{1}[-\dim(V)] \qquad \square$$

Lax Symmetric monoidality of homology

Observation. Let $M \in \mathrm{CMon}(\mathrm{LCH})$. Then the slice category $\mathrm{LCH}_{/M}$ admits a natural symmetric monoidal structure $\otimes$ with

$$(X,f) \otimes (Y,g) := (X \times Y, f+g)$$

where $f+g : X \times Y \longrightarrow M$ is the composite

$$X \times Y \xrightarrow{f \times g} M \times M \xrightarrow{\mathrm{add}} M.$$

> The following is a result that is quite techical to prove, but intuitively clear.

Thm. Let $M \in \mathrm{CMon}(\mathrm{LCH})$. There is a lax symmetric monoidal functo

$$\Gamma_M : (\mathrm{LCH}_{/M}, \otimes) \longrightarrow (\mathrm{Sh}(M), *)$$

given on objects by

$$(X,f) \longmapsto f_! f^! (\omega_M).$$

homology of X over M with coeffs in ω_M.

The assignment on morphisms sends

$$\begin{tikzcd} X \arrow[rr, "p"] \arrow[dr, "f"'] & & Y \arrow[dl, "g"] \\ & M & \end{tikzcd}$$

to

$$f_! f^!(\omega_M) \simeq g_!(p_! p^!) g^!(\omega_M) \xrightarrow{\text{counit}} g_! g^!(\omega_M)$$

Observation Let N be a lattice and $X, Y \subset M_{\mathbb{R}}$ be open or closed subspaces. Then we have a commutative triangle

$$\begin{tikzcd} X \times Y \arrow[rr, "p"] \arrow[dr, "a"'] & & X + Y \arrow[dl] \\ & M_{\mathbb{R}} & \end{tikzcd}$$

where the source is the tensor product of the inclusions $X \hookrightarrow M_{\mathbb{R}}$ and $Y \hookrightarrow M_{\mathbb{R}}$, the target is the inclusion of the

Minkowski sum, and p is the addition map. Thus by the lax symmetric monoidality of $\Gamma_{M_{\mathbb{R}}}$, we have a natural map

$$\Gamma_{M_{\mathbb{R}}}(X) * \Gamma_{M_{\mathbb{R}}}(Y) \longrightarrow \Gamma_{M_{\mathbb{R}}}(X \times Y) \xrightarrow{\Gamma_{M_{\mathbb{R}}}(p)} \Gamma_{M_{\mathbb{R}}}(X+Y)$$

lax sym.
mon structure

The homology of a cone

Recall. For N a lattice, $\mathrm{Poly}(M)$ is the poset of subsets of $M_{\mathbb{R}}$ of the form (cone) + (polytope). For $\sigma \subset N_{\mathbb{R}}$ a cone, we write

$$\Theta(\sigma) = \{ m + \sigma^\vee \mid m \in M \}$$
$$\subset \mathrm{Mod}_{\sigma^\vee} \mathrm{Poly}(M)$$

> We'll start by analyzing $\Gamma_{M_{\mathbb{R}}} : \mathrm{Poly}(M) \longrightarrow \mathrm{Sh}(M_{\mathbb{R}})$.

Prop. Let N be a lattice and let X and Y be open or closed polyhedral subsets of $M_{\mathbb{R}}$ of maximal dimension. Then the natural map

$$\Gamma_{M_{\mathbb{R}}}(X) * \Gamma_{M_{\mathbb{R}}}(Y) \longrightarrow \Gamma_{M_{\mathbb{R}}}(X+Y)$$

is an equivalence.

> First we state some corollaries

Cor. Let N be a lattice and $\sigma \subset N_{\mathbb{R}}$ a cone. Then the sheaf

$$\omega_{\sigma^\vee} \simeq \Gamma_{M_{\mathbb{R}}}(\sigma^\vee)$$

on $M_{\mathbb{R}}$ is an idempotent algebra with respect to convolution.

Proof

Since $\sigma^\vee$ is an idempotent algebra with respect to Minkowski sum and is of maximal dimension, we deduce that

$$\Gamma_{M_{\mathbb{R}}}(\sigma^\vee) * \Gamma_{M_{\mathbb{R}}}(\sigma^\vee) \xrightarrow{\sim} \Gamma_{M_{\mathbb{R}}}(\sigma^\vee + \sigma^\vee) \simeq \Gamma_{M_{\mathbb{R}}}(\sigma^\vee) \qquad \square$$

Cor. Let N be a lattice. Then the assignment $\sigma \mapsto \omega_{\sigma^\vee}$ defines a functor

$$\mathrm{Cone}(N_{\mathbb{R}})^{op} \longrightarrow \mathrm{CAlg}^{idem}(\mathrm{Sh}(M_{\mathbb{R}}), *)$$

Proof.

Recall that for a symmetric monoidal ∞-category $\mathcal{C}$, the full subcategory $\mathrm{CAlg}^{\mathrm{idem}}(\mathcal{C}) \subset \mathrm{CAlg}(\mathcal{C})$ is a poset, and there is a map of idem algebras $A \longrightarrow B$ if and only if B is an A-algebra. So the functoriality can be specified at the 1-categorical level, which is obvious $\square$

> For the proof of the prop we need

Lemma Let N be a lattice and $\bar{U} \subset M_{\mathbb{R}}$ be a closed polyhedral subset of top dimension with interior U. Then the map of homology sheaves

$$\Gamma_{M_{\mathbb{R}}}(U) \longrightarrow \Gamma_{M_{\mathbb{R}}}(\bar{U})$$

is an equivalence

Proof of Lemma

Since every sheaf on $M_{\mathbb{R}}$ is a hypersheaf, it suffices to check that this map induces an equivalence on stalks. If $x \notin \partial \bar{U}$, this is clear by proper base change: both agree with the restriction to U on U and are 0 outside of $\bar{U}$

For $x \in \partial \bar{u}$, the stalk of the left-hand side vanishes. So we just need to check this for the right-hand side. To compute the stalk, let $(D_k)_{k \geq 0}$ be disks of shrinking radii converging to 0 centered at x. Then

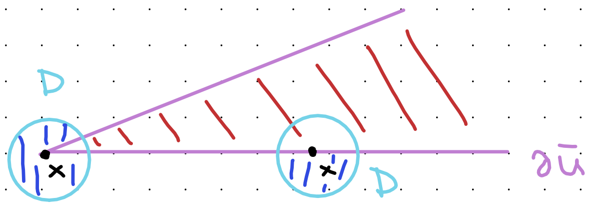
$$\Gamma_{M_{\mathbb{R}}}(\bar{u})_x \simeq \operatorname*{Colim}_{k \to \infty} \Gamma_{M_{\mathbb{R}}}(\bar{u})(D_k)$$

Using that $\omega_{M_{\mathbb{R}}} \simeq \mathbb{1}[n]$ for $n = \dim(M_{\mathbb{R}})$ and applying proper basechange we see that for $i : \bar{u} \hookrightarrow M_{\mathbb{R}}$ the inclusion, we have

$$\Gamma_{M_{\mathbb{R}}}(\bar{u})(D_k) \simeq (i_! i^! (\mathbb{1}[n]))(D_k)$$

$$\simeq \operatorname{fib}(\mathbb{1}(D_k) \longrightarrow \mathbb{1}(D_k - \bar{u}))[n].$$

Since $\bar{u}$ is polyhedral, for a sufficiently small ball D, $D \setminus \bar{u}$ is also contractible. Here's the picture explaining why this is true:



So if $k \gg 0$, then by the recognition criterion for locally constant sheaves, the restriction map

$$\mathbb{1}(D_k) \longrightarrow \mathbb{1}(D_k - \bar{u})$$

is an equivalence So its fiber is zero. Hence

$$\Gamma_{M_{\mathbb{R}}}(\bar{u})_x \simeq \operatorname*{Colim}_{k \gg 0} \Gamma_{M_{\mathbb{R}}}(\bar{u})(D_k)$$

$$\simeq \operatorname*{Colim}_{k \gg 0} 0 = 0,$$

as desired. $\square$

Proof of convolution formula for $\Gamma_{M_{\mathbb{R}}}$:

Note that by the naturality of the comparison map

$$\Gamma_{M_{\mathbb{R}}}(X) * \Gamma_{M_{\mathbb{R}}}(Y) \longrightarrow \Gamma_{M_{\mathbb{R}}}(X+Y)$$

and the previous lemma, it suffices to treat the case where X and Y are open. This computation will reduce to the computation from the beginning of the lecture.

We'll cheat a little bit and only show both sides are abstractly equivalent, but being a little more careful proves the claim. Let $n = \dim(M_{\mathbb{R}})$. Note that since X, Y, and $X+Y$ are all homeomorphic to $\mathbb{R}^n$, their dualizing sheaves are $\mathbb{1}_X[n]$, $\mathbb{1}_Y[n]$, and $\mathbb{1}_{X+Y}[n]$. Hence we compute

$$\begin{aligned}
\omega_X * \omega_Y &\simeq (\mathbb{1}_X[n]) * (\mathbb{1}_Y[n]) \\
&\simeq (\mathbb{1}_X * \mathbb{1}_Y)[2n] \\
&\simeq (\mathbb{1}_{X+Y}[-n])[2n] \quad \text{from earlier} \\
&\simeq \mathbb{1}_{X+Y}[n] \\
&\simeq \omega_{X+Y}.
\end{aligned}$$

$\square$

Combinatorial to constructible comparison

Observe If (N, Σ) is a fan, then the assignment

$$\sigma \mapsto \mathrm{Mod}_{\omega_{\sigma^\vee}} \mathrm{Sh}(M_{\mathbb{R}})$$

with functoriality given by tensoring defines a diagram

$$\Sigma^{op} \longrightarrow \mathrm{CAlg}(\mathrm{Pr}^{L})$$

Taking limits, we obtain a symmetric monoidal left adjoint

$$L_\Sigma : \mathrm{Sh}(M_{\mathbb{R}}) \longrightarrow \lim_{\sigma \in \Sigma^{op}} \mathrm{Mod}_{\omega_{\sigma^\vee}} \mathrm{Sh}(M_{\mathbb{R}}).$$

Write R_Σ for the lax symmetric monoidal right adjoint to L_Σ, induced by the fully faithful forgetful functors

$$\mathrm{Mod}_{\omega_{\sigma^\vee}} \mathrm{Sh}(M_{\mathbb{R}}) \longrightarrow \mathrm{Sh}(M_{\mathbb{R}})$$

Since each of these functors is fully faithful, R is fully faithful.

Construction Let N be a lattice and $\sigma \subset N_{\mathbb{R}}$ a cone. Write φ_σ for the composite

$$\Theta(\sigma) \longrightarrow \mathrm{Mod}_{\sigma^\vee}\mathrm{Poly}(M) \xrightarrow{\Gamma_{M_{\mathbb{R}}}} \mathrm{Mod}_{\omega_{\sigma^\vee}}\mathrm{Sh}(M_{\mathbb{R}})$$

Recall that the first functor is symmetric monoidal, and the second is lax symmetric monoidal

Observe The functor $\varphi_\sigma : \Theta(\sigma) \longrightarrow \mathrm{Mod}_{\omega_{\sigma^\vee}}\mathrm{Sh}(M_{\mathbb{R}})$ is sym monoidal

> This is because every element of $\Theta(\sigma)$ is of maximal dimension, and we showed that for such $X, Y \in \mathrm{Poly}(M)$, we have
$$\Gamma_{M_{\mathbb{R}}}(X) * \Gamma_{M_{\mathbb{R}}}(Y) \xrightarrow{\sim} \Gamma_{M_{\mathbb{R}}}(X+Y).$$

Using the universal property of Day convolution, we write

$$\Psi_\sigma : (\mathrm{Fun}(\Theta(\sigma), \mathrm{Sp}), \otimes_{\mathrm{Day}}) \longrightarrow (\mathrm{Mod}_{\omega_{\sigma^\vee}}\mathrm{Sh}(M_{\mathbb{R}}), *)$$

for the symmetric monoidal left adjoint induced by φ_σ.

Check (slightly technical). Given a fan (N,Σ), the functors $(\Psi_\sigma)_{\sigma\in\Sigma^{op}}$ assemble into a natural transformation

$$\mathrm{Fun}(\Theta(-)^{op},\mathrm{Sp}) \longrightarrow \mathrm{Mod}_{\omega_{(-)^\vee}}\mathrm{Sh}(M_{\mathbb{R}})$$

of diagrams $\Sigma^{op}\longrightarrow \mathrm{CAlg}(\mathrm{Pr}^L)$.

Notation. For a fan (N,Σ), write

$$\Psi_\Sigma \quad \lim_{\sigma\in\Sigma^{op}} \mathrm{Fun}(\Theta(\sigma)^{op},\mathrm{Sp}) \longrightarrow \lim_{\sigma\in\Sigma^{op}} \mathrm{Mod}_{\omega_{\sigma^\vee}}\mathrm{Sh}(M_{\mathbb{R}})$$

for the induced symmetric monoidal functor.

Def. Let (N,Σ) be a fan. The **coherent-constructible correspondence**

$$\kappa_\Sigma\ (\mathrm{QCoh}(X_\Sigma/T),\otimes) \longrightarrow (\mathrm{Sh}(M_{\mathbb{R}}),\ast)$$

is the lax symmetric monoidal functor given by the composite

$$
\begin{tikzcd}
QCoh(X_\Sigma/T) \arrow[r, "\Phi_\Sigma^{-1}", "\sim"'] & \lim\limits_{\sigma\in\Sigma^{op}} Fun(\Theta(\sigma)^{op}, Sp) \arrow[r] & \lim\limits_{\sigma\in\Sigma^{op}} Mod_{\omega_\sigma^\vee} Sh(M_{\mathbb{R}}) \arrow[d, "R_\Sigma"] \\
& & Sh(M_{\mathbb{R}})
\end{tikzcd}
$$

inverse of equivalence
we constructed earlier

Note that the only functor here which we have not seen is symmetric monoidal is R_Σ

Next time When Σ is smooth projective, R_Σ is an equivalence, so k_Σ is a symmetric monoidal left adjoint